



Preface

About the Book

If data is the new oil, then machine learning is the drill. As companies gain access to ever-increasing quantities of raw data, the ability to deliver state-of-the-art predictive models that support business decision-making becomes more and more valuable.

In this book, you'll work on an end-to-end project based around a realistic data set and split up into bite-sized practical exercises. This creates a case-study approach that simulates the working conditions you'll experience in real-world data science projects.

You'll learn how to use key Python packages, including pandas, Matplotlib, and scikit-learn, and master the process of data exploration and data processing, before moving on to fitting, evaluating, and tuning algorithms such as regularized logistic regression and random forest.

Now in its second edition, this book will take you through the end-to-end process of exploring data and delivering machine learning models. Updated for 2021, this edition includes brand new content on XGBoost, SHAP values, algorithmic fairness, and the ethical concerns of deploying a model in the real world.

By the end of this data science book, you'll have the skills, understanding, and confidence to build your own machine learning models and gain insights from real data.

About the Author

Stephen Klosterman is a Machine Learning Data Scientist with a background in math, environmental science, and ecology. His education includes a Ph.D. in Biology from Harvard University, where he was an assistant teacher of the Data Science course. His professional experience includes work in the environmental, health care, and financial sectors. At work, he likes to research and develop machine learning solutions that create value, and that stakeholders understand. In his spare time, he enjoys running, biking, paddleboarding, and music.

Objectives

- Load, explore, and process data using the pandas Python package
- Use Matplotlib to create effective data visualizations
- Implement predictive machine learning models with scikit-learn and XGBoost
- Use lasso and ridge regression to reduce model overfitting
- Build ensemble models of decision trees, using random forest and gradient boosting
- Evaluate model performance and interpret model predictions
- Deliver valuable insights by making clear business recommendations

Audience

Data Science Projects with Python – Second Edition is for anyone who wants to get started with data science and machine learning. If you're keen to advance your career by using data analysis and predictive modeling to generate business insights, then this book is the perfect place to begin. To quickly grasp the concepts covered, it is recommended that you have basic experience with programming in Python or another similar language (R, Matlab, C, etc). Additionally, knowledge of statistics that would be covered in a basic course, including topics such as probability and linear regression, or a willingness to

learn about these on your own while reading this book would be useful.

Approach

Data Science Projects with Python takes a practical case study approach to learning, teaching concepts in the context of a real-world dataset. Clear explanations will deepen your knowledge, while engaging exercises and challenging activities will reinforce it with hands-on practice.

About the Chapters

Chapter 1, Data Exploration and Cleaning, gets you started with Python and Jupyter notebooks. The chapter then explores the case study dataset and delves into exploratory data analysis, quality assurance, and data cleaning using pandas.

Chapter 2, Introduction to Scikit-Learn and Model Evaluation, introduces you to the evaluation metrics for binary classification models. You'll learn how to build and evaluate binary classification models using scikit-learn.

Chapter 3, Details of Logistic Regression and Feature Exploration, dives deep into logistic regression and feature exploration. You'll learn how to generate correlation plots of many features and a response variable and interpret logistic regression as a linear model.

Chapter 4, The Bias-Variance Trade-Off, explores the foundational machine learning concepts of overfitting, underfitting, and the bias-variance trade-off by examining how the logistic regression model can be extended to address the overfitting problem.

Chapter 5, Decision Trees and Random Forests, introduces you to tree-based machine learning models. You'll learn how to train decision

trees for machine learning purposes, visualize trained decision trees, and train random forests and visualize the results.

Chapter 6, Gradient Boosting, XGBoost, and SHAP Values, introduces you to two key concepts: gradient boosting and **shapley additive explanations (SHAP)**. You'll learn to train XGBoost models and understand how SHAP values can be used to provide individualized explanations for model predictions from any dataset.

Chapter 7, Test Set Analysis, Financial Insights, and Delivery to the Client, presents several techniques for analyzing a model test set for deriving insights into likely model performance in the future. The chapter also describes key elements to consider when delivering and deploying a model, such as the format of delivery and ways to monitor the model as it is being used.

Hardware Requirements

For the optimal student experience, we recommend the following hardware configuration:

- Processor: Intel Core i5 or equivalent
- Memory: 4 GB RAM
- Storage: 35 GB available space

Software Requirements

You'll also need the following software installed in advance:

- OS: Windows 7 SP1 64-bit, Windows 8.1 64-bit or Windows 10 64-bit, Ubuntu Linux, or the latest version of OS X
- Browser: Google Chrome/Mozilla Firefox Latest Version
- Notepad++/Sublime Text as IDE (this is optional, as you can practice everything using the Jupyter Notebook on your browser)

- Python 3.8+ (This book uses Python 3.8.2) installed (from <https://python.org>, or via Anaconda as recommended below) . At the time of writing, the SHAP library used in *Chapter 6, Gradient Boosting, XGBoost, and SHAP Values*, is not compatible with Python 3.9. Hence, if you are using Python 3.9 as your base environment, we suggest that you set up a Python 3.8 environment as described in the next section.
- Python libraries as needed (Jupyter, NumPy, Pandas, Matplotlib, and so on, installed via Anaconda as recommended below)

Installation and Setup

Before you start this book, it is recommended to install the Anaconda package manager and use it to coordinate installation of Python and its packages.

Code Bundle

Please find the code bundle for this book, hosted on GitHub at <https://github.com/PacktPublishing/Data-Science-Projects-with-Python-Second-Ed>.

Anaconda and Setting up Your Environment

You can install Anaconda by visiting the following link: <https://www.anaconda.com/products/individual>. Scroll down to the bottom of the page and download the installer relevant to your system.

It is recommended to create an environment in Anaconda to do the exercises and activities in this book, which have been tested against the software versions indicated here. Once you have Anaconda installed, open a Terminal, if you're using macOS or Linux, or a Command Prompt window in Windows, and do the following:

1. Create an environment with most required packages. You can call it whatever you want; here it's called **dspwp2**. Copy and paste, or type the entire statement here on one line in the terminal:

```
conda create -n dspwp2 python=3.8.2 jupyter=1.0.0
pandas=1.2.1 scikit-learn=0.23.2 numpy=1.19.2
matplotlib=3.3.2 seaborn=0.11.1 python-graphviz=0.15
xlrd=2.0.1
```

2. Type 'y' and press [Enter] when prompted.
3. Activate the environment:

```
conda activate dspwp2
```

4. Install the remaining packages:

```
conda install -c conda-forge xgboost=1.3.0
shap=0.37.0
```

5. Type 'y' and [Enter] when prompted.
6. You are ready to use the environment. To deactivate it when finished:

```
conda deactivate
```

We also have other code bundles from our rich catalog of books and videos available at <https://github.com/PacktPublishing/>. Check them out!

Conventions

Code words in the text, database table names, folder names, filenames, file extensions, pathnames, dummy URLs, user input, and Twitter handles are shown as follows: "By typing **conda list** at the command line, you can see all the packages installed in your environment."

A block of code is set as follows:

```
import numpy as np #numerical computation
import pandas as pd #data wrangling
import matplotlib.pyplot as plt #plotting package
#Next line helps with rendering plots
```

```
%matplotlib inline
import matplotlib as mpl #add'l plotting functionality
mpl.rcParams['figure.dpi'] = 400 #high res figures
import graphviz #to visualize decision trees
```

New terms and important words are shown in bold. Words that you see on the screen, for example, in menus or dialog boxes, appear in the text like this: "Create a new Python 3 notebook from the **New** menu as shown."

Code Presentation

Lines of code that span multiple lines are split using a backslash (\). When the code is executed, Python will ignore the backslash, and treat the code on the next line as a direct continuation of the current line.

For example:

```
my_new_lr = LogisticRegression(penalty='l2',
                                dual=False,\
                                tol=0.0001, C=1.0,\
                                fit_intercept=True,\
                                intercept_scaling=1,\
                                class_weight=None,\
                                random_state=None,\
                                solver='lbfgs',\
                                max_iter=100,\
                                multi_class='auto',\
                                verbose=0,
                                warm_start=False,\
                                n_jobs=None,
                                l1_ratio=None)
```

Comments are added into code to help explain specific bits of logic. Single-line comments are denoted using the # symbol, as follows:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt #import plotting
package
#render plotting automatically
%matplotlib inline
```

Get in Touch

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